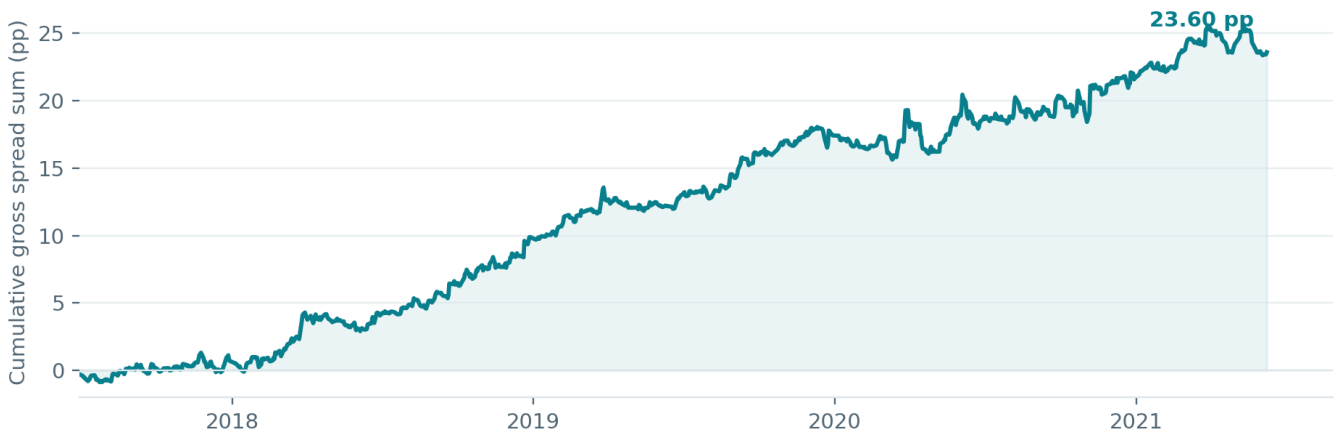
01 / EXECUTIVE RESULT

Positive spreads, modest signal

Completed experiment: 8 folds x 20 epochs. Validation Target dates: 30 Jun 2017 to 08 Jun 2021.

POOLED SHARPE	MEAN RANK IC	GROSS SPREAD SUM
0.0994	0.00375	23.60 pp
Daily; non-annualized	957 defined daily values	Additive; before costs

958 DATES / 1,850,796 VALIDATION SAMPLES



Original daily spreads added over time, using +0.5 long / -0.5 short exposure. This curve is not compounded NAV.

What the completed run supports

Every fold has a positive gross spread sum. However, the mean daily rank correlation is small, and folds 3 and 5 provide 50.98% of the total. The ranking signal varies materially across periods.

What it does not establish

Validation MAE is 0.0160325 versus 0.0159500 for predicting zero: the model error is 0.52% higher. These results do not establish a robust, cost-adjusted trading advantage or a Kaggle leaderboard score.

02 / METHOD AND JPX CONTEXT

The original experiment, preserved

One model and one Adam state continue through all folds. There is no restart between folds.

Component	Recorded setup
Model	Conv1d: 6 to 64 channels, kernel 3; ReLU; max pool 2; 3-layer LSTM, hidden size 100; linear output. Dropout 0.2.
Inputs	42 valid observations per security; Open, High, Low, Close, Volume and ExpectedDividend.
Training	20 epochs per fold; batch 64; shuffle=False; L1 loss; Adam, learning rate 0.001. Seed 20261016.
Validation	8 expanding-window folds with 180-calendar-day intervals. Remove the last two available training dates before fitting the model.
Preprocessing	Fit scaling on the pre-purge training partition; reuse it in validation. Add up to 41 earlier input rows for validation windows.
Coverage	958 dates and 1,850,796 samples after original filtering. Trailing partial split is not evaluated.
Execution	Cloud CPU, 8 PyTorch threads. Completed-epoch recovery preserved model, Adam, random states and saved evaluations.

Primary JPX comparison

JPX scores the Sharpe ratio of daily rank-weighted spreads from the top 200 and bottom 200 securities. The report therefore leads with the original non-annualized Sharpe; Rank IC and MAE are supporting diagnostics. [1,2]

Offline result, not a submitted score

The project uses +0.5 long and -0.5 short weights. Its internal average ranks for ties are not a valid submission column by themselves. Universe coverage, unique zero-based ranks and API row order still need end-to-end submission checks. [2,4]

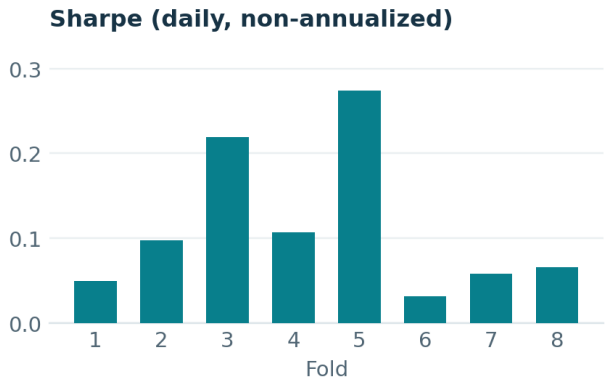
03 / FOLD-LEVEL EVALUATION

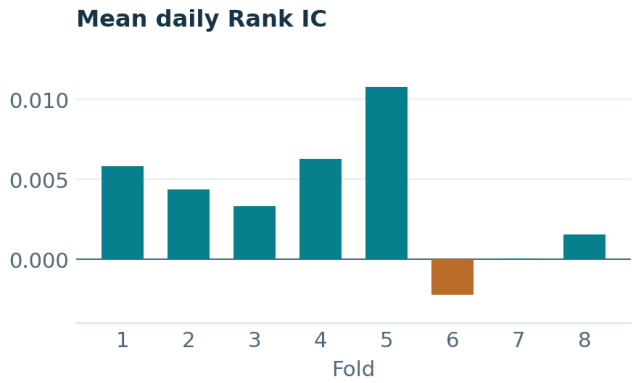
All eight folds completed

Each row includes 20 completed epochs and its full validation. Sharpe is daily and non-annualized.

Fold	Target date range	Days	MAE	Sharpe	Rank IC	Gross sum (pp)
1	2017-06-30 2017-12-26	122	0.012257	0.0492	0.00580	0.997
2	2017-12-27 2018-06-22	119	0.014805	0.0974	0.00437	2.532
3	2018-06-25 2018-12-21	125	0.017365	0.2196	0.00330	6.080
4	2018-12-25 2019-06-19	114	0.015769	0.1067	0.00629	2.375
5	2019-06-20 2019-12-16	121	0.013416	0.2744	0.01080	5.950
6	2019-12-17 2020-06-12	117	0.022397	0.0318	-0.00224	1.279
7	2020-06-15 2020-12-10	121	0.016862	0.0583	0.00003	2.443
8	2020-12-11 2021-06-08	119	0.015345	0.0662	0.00156	1.940

Gross sum = sum of original daily spread returns, before costs. Fold values preserve the original saved calculations.





Strongest fold Sharpe: 5 (0.2744). Weakest: 6 (0.0318), which also has negative Rank IC. The pooled score is computed from all daily returns, not the mean of these eight Sharpes.

04 / DIAGNOSTIC INTERPRETATION

MAE trails the zero baseline

The model can select useful extremes while showing little improvement in all-stock prediction error.

MODEL VALIDATION MAE

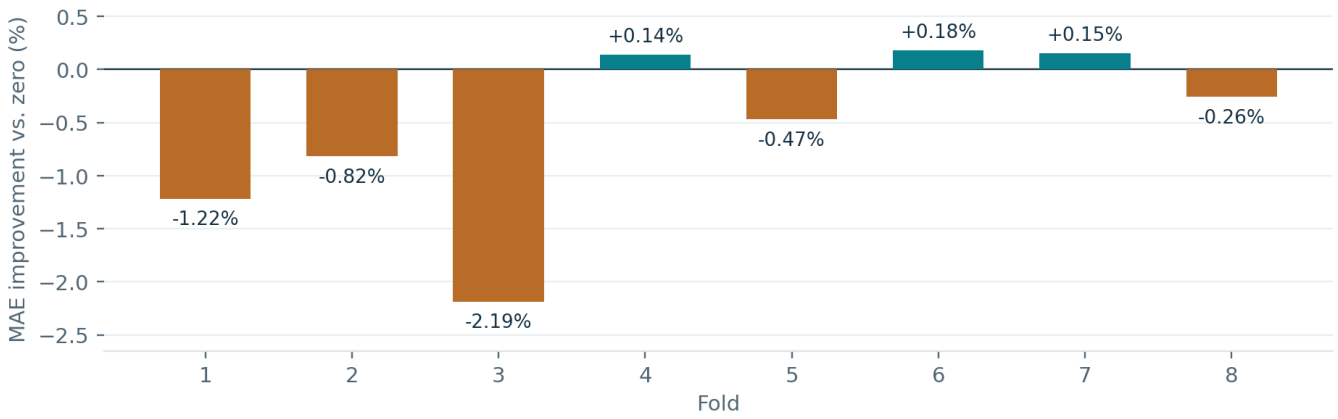
0.0160325

Sample-weighted across all folds

ZERO-PREDICTION MAE

0.0159500

Model error is 0.52% higher



Positive bars mean lower MAE than the zero baseline. The zero baseline is a prediction-error comparator, not a ranking strategy.

Performance is concentrated and uneven

Folds 3 and 5 contribute 50.98% of the gross spread sum. The five strongest dates contribute 30.07%. Positive spreads occur on 52.61% of dates and in 35 of 49 calendar months. These are descriptive counts, not significance tests.

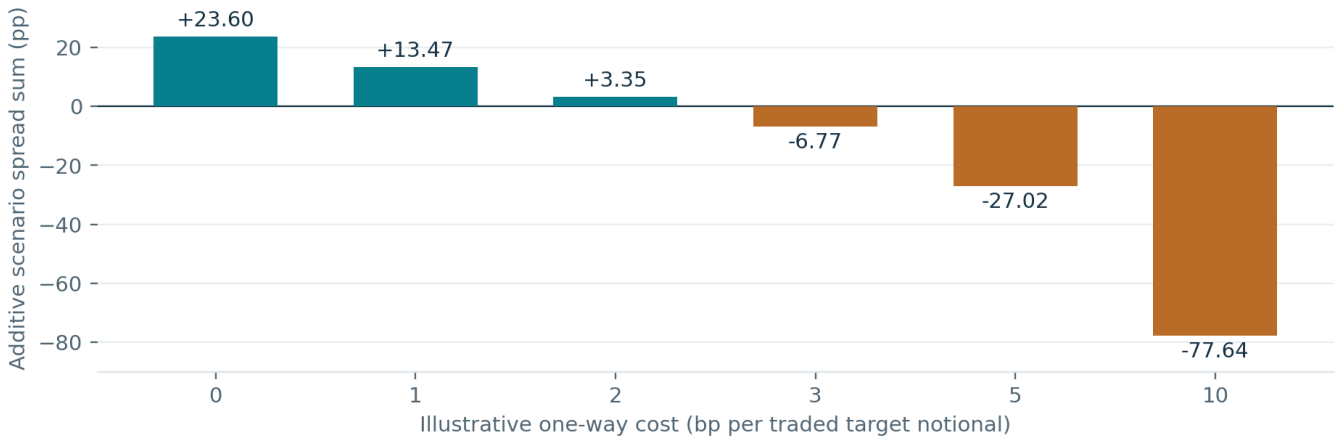
One missing IC value has a data explanation

On 29 Sep 2020 the supplied Targets are all zero, so Spearman correlation is undefined. The 1,973 evaluated securities have 1,973 distinct predictions. The original mean skips this date; this is not constant model output.

05 / SECONDARY COST ANALYSIS

A small cushion under simple costs

Illustrative sensitivity only. Costs were not part of the original reported JPX-style score.



Break-even cost: 2.33 bp per traded target notional

Subtract cost x sum of absolute daily target-weight changes, starting and ending flat. At 2 bp the additive spread sum remains +3.35 pp; at 3 bp it becomes -6.77 pp. This simplified result is not realized net performance.

The scenario excludes price drift, marked-to-market rebalancing, execution, liquidity limits, short borrow and taxes. The mean absolute target-weight change is 1.0567 per date, including the initial and final trades.

READ THE UNITS CORRECTLY

Term	Meaning in this report
bp / bps	Basis point(s): 1 bp = 0.01% = 0.0001 in decimal-return units.
pp	Percentage point(s). A gross spread sum of 0.23595 is shown as 23.595 pp, not a compounded investor gain.
Gross spread sum	Daily long-short returns added before costs. The additive maximum peak-to-trough fall is 3.23 pp, not NAV drawdown.

06 / EVIDENCE AND NEXT EXPERIMENTS

Complete, traceable, and bounded

Analysis is based on frozen predictions. Report preparation did not retrain or change the algorithm.

Evidence	Recorded result
Completed	17 Oct 2026, 03:35:42 UTC (11:35:42 in Taipei).
Total wall time	18 h 25 m 53 s, including interruptions and downtime.
Recorded training	11 h 00 m 57 s summed over 160 completed epochs; excludes preprocessing, validation, interrupted partial work and downtime.
Completion checks	8 fold evaluations; finite predictions; final model/normalization match; continued Adam state with 2,373,700 cumulative steps.

Research limits and sensible next tests

This is one seed and one continuous historical training path; the folds are not independent trials. Next tests should use an untouched period, multiple seeds and simple ranking baselines, followed by an exact competition-API evaluation. These tests are not included in the present results.

The square-root-of-252 Sharpe reference is 1.5787, under an IID/no-serial-correlation convention. Lag-1 daily return autocorrelation is 0.1273. The primary score remains 0.0994, non-annualized; the reference is not an autocorrelation-adjusted estimate. [3]

SOURCES AND SUPPORTING FILES

[1] Kaggle JPX evaluation overview

[2] Kaggle JPX code requirements and data description

[3] A. W. Lo (2002), The Statistics of Sharpe Ratios

[4] Frozen project source and original evaluation

Full URLs and evaluation formulae are in details/evaluation_protocol.md. Exact parameters, per-fold and daily data, original artifact hashes, charts and reproduction scripts are in details/. Public competition descriptions accessed 17 Oct 2026.